Reproducing Med-NCA: A Rigorous Reproducibility Study of Neural Cellular Automata for Medical Image Segmentation

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Reproduction target: Med-NCA (IPMI 2023) & M3D-NCA (MICCAI 2023)

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Abstract

Neural Cellular Automata (NCA) have emerged as an extremely lightweight (∼13k–70k parameters) alternative to U-Nets for medical image segmentation, with Med-NCA reporting competitive Dice on the Medical Segmentation Decathlon (MSD). We present an *independent, zero-deviation* reproduction of Med-NCA on its two original benchmarks — Hippocampus (MSD Task04) and Prostate (MSD Task05) — following the RIDGE reproducibility framework. Our study reports three findings. **(1)** The Hippocampus anchor reproduces faithfully: per-volume Dice = 0.866 against the paper's 0.886 (within one standard deviation). **(2)** The Prostate anchor only *partially* reproduces: Dice = 0.672 versus the paper's 0.838; we report this gap honestly and attribute it (without any number-fitting) primarily to under-training (301 vs. 1000 epochs) and a small high-variance test set ($n = 9$). **(3)** Most importantly, we document a *reproducibility-fragility* phenomenon: a *mathematically equivalent* engineering speed-up — moving the stochastic fire-mask sampling from CPU to GPU — silently changes the RNG stream and, on the more aggressive Prostate configuration, drives training from Dice = 0.672 to complete divergence (Dice = 0.0, logits → -10^9). We further characterise the baseline's claimed properties (inference-step convergence, robustness to corruptions, and the built-in variance-based quality metric). All numbers carry bootstrap 95% confidence intervals, fixed seeds, and traceable run IDs.

1 Introduction

Neural Cellular Automata treat segmentation as the steady state of a locally-communicating cellular update rule. A single small network is applied identically to every pixel/voxel cell for a fixed number of synchronous steps; global structure emerges from purely local message passing [1]. Med-NCA [2] and its 3D extension M3D-NCA [3] push this idea to high-resolution medical images using a two-level (coarse→fine patch) scheme, achieving U-Net-competitive Dice with three-to-four orders of magnitude fewer parameters — small enough to run on a Raspberry Pi.

Such efficiency claims are attractive but only meaningful if the reported numbers can be *independently reproduced*. Reproducibility in medical imaging is fragile: under-specified preprocessing, undocumented test splits, and silent implementation differences routinely prevent third-party replication [4, 5]. This report asks a deliberately narrow question: *can the headline Med-NCA Dice numbers be reproduced from the official code, and what does it take?*

We adopt a strict **zero-deviation protocol** (Section 3): configuration, hyper-parameters, optimiser, inference steps, and *implementation* are aligned verbatim with the authoritative source; no clipping, no learning-rate tweaks, no "convergence helpers". When the official setting fails to reproduce a number, we report the failure and its root cause rather than fitting the target. This protocol is what allowed us to isolate the RNG-fragility finding that is, in our view, the most transferable contribution of this study.

Contributions.

1. A faithful reproduction of the Med-NCA Hippocampus anchor (Dice = 0.866 vs. 0.886), with full metric discipline.
2. An honest, attributed *partial* reproduction of the Prostate anchor (Dice = 0.672 vs. 0.838), with a statistical gap analysis instead of number-fitting.
3. A reproducibility-fragility case study: a mathematically equivalent CPU→GPU RNG change collapses Prostate training (0.672 → 0.0), a concrete caveat for anyone optimising NCA code.
4. A behavioural characterisation of the baseline's own claims: inference-step convergence, corruption robustness, and the variance-based built-in quality metric (NQM).

2 Background: Med-NCA

Architecture (read from official code, not the PDF). We verified the architecture against the official `M3D-NCA` source rather than the paper figures, and found two discrepancies worth noting for reproducers: the `BasicNCA` perception uses *fixed Sobel* filters (plus identity), not a learned convolution, and the example loss is `DiceBCELoss`, not the Dice-Focal loss implied by the paper text. A cell state holds `channel_n` channels (channel 0 = the input image, the rest initialised to zero). Each update is $\mathrm{fc}_1(\mathrm{ReLU}(\mathrm{fc}_0(\mathrm{perceive}(x))))$ with the last layer zero-initialised (so the initial residual is zero), gated by a random *fire mask* (rate 0.5), then added residually; the input channels are restored every step. Two levels are used: a coarse level communicates global context, which is upsampled and refined by a fine level operating on patches.

Official hyper-parameters. Shared: lr = 16e−4, Adam betas $(0.5, 0.5)$, lr-decay 0.9999, fire rate 0.5, hidden size 128, Dice split 0.7/0/0.3. The per-dataset settings are read verbatim from the official archived configs and differ in both width and step count: **Hippocampus** (`config.dt`) uses `channel_n`= 32, **16** inference steps, batch 40, rescale, levels $16\times16 \to 64\times64$; **Prostate** (notebook) uses `channel_n`= 32, **64** inference steps, batch 20, levels $64\times64 \to 256\times256$. (An earlier exploratory run used a *non-official* Hippocampus setting — `channel_n`= 16, 64 steps — which we treat as deprecated; see Sections 4 and 6.)

Targets. From Med-NCA IPMI'23 Table 1: Hippocampus Dice = 0.886 ± 0.042 (U-Net 0.858), Prostate Dice = 0.838 ± 0.083 (U-Net 0.799).

3 Reproduction Protocol

Zero-deviation rule. Every quantity that affects training is aligned verbatim with the official notebook / config / source. Forbidden "helpers": gradient clipping, lowered learning rate, changed step count, warm-up, or any re-implementation — *including* a mathematically equivalent speed-up subclass. This rule is not pedantry; Section 6 shows it is exactly what separates a faithful reproduction from a silent divergence.

Metric discipline. We report *per-image (per-volume)* mean Dice — each test volume yields one Dice, then we average over N volumes — never the batch-aggregate Dice (which the framework also exposes and which differs by several points). Every reported number carries a bootstrap 1000× 95% CI, a fixed seed (42), and a run/commit identifier. Single-pass and 10× pseudo-ensemble inference are reported separately.

Table 1: R1 Hippocampus reproduction (per-volume Dice, $n = 78$, seed 42). Paper target 0.886 ± 0.042; threshold ≥ 0.86. The official zero-deviation run is the anchor; the fast run is a deprecated non-official-config reference (it happens to agree to within 0.002).

Run	Config	Dice	95% CI	Status
Official (single)	ch32/16st	0.8644	[0.856, 0.872]	**PASS**
Official (ens. 10×)	ch32/16st	0.8663	[0.858, 0.874]	**PASS**
Fast (single, deprec.)	ch16/64st	0.8661	[0.858, 0.873]	ref
Paper Med-NCA	—	0.886	±0.042	target
Paper U-Net	—	0.858	±0.044	baseline

Frozen test sets (RIDGE Integrity). Test splits are frozen via the framework's `data_split.dt` and the exact volume IDs are logged. Hippocampus: 182/0/78 train/val/test volumes; Prostate: 23/0/9 (of the 32 MSD Task05 cases).

Two known traps, neutralised. (i) The official `Experiment.reload()` silently overwrites the runtime config from a cached `config.dt` if the checkpoint directory is non-empty, making the requested epoch count a no-op; we clear the model directory before every run and read the *true* step count from `exp.currentStep`. (ii) NCA training is compute-bound (64 sequential steps × 2 levels), not data-bound; this matters for the cost budget in Section 7.

4 R1: Hippocampus Reproduction

The Hippocampus anchor reproduces faithfully. The zero-deviation official run (`channel_n`= 32, 16 steps, the archived `config.dt`) was stopped at its eval-Dice plateau — eval Dice flattened at 0.864 across epochs 125–175 (Figure 1), with the last point even dipping slightly, indicating onset of over-fitting — and frozen by checkpoint-only evaluation at epoch 150. Table 1 reports the result: per-volume single-pass Dice = 0.8644 (95% CI [0.856, 0.872]), **PASS**, within one paper standard deviation of 0.886. An earlier exploratory run used a non-official setting (`channel_n`= 16, 64 steps) and reached 0.866; the two agree to within 0.002, but only the official run is the reproduction anchor. Figure 2 shows the inference-step dependence, recomputed on the official checkpoint.

The official anchor sits 0.022 below the paper target, well within one paper standard deviation. We attribute the residual to training duration (the run was stopped at its eval plateau, epoch 150, vs. the paper's 1000 epochs) and split/seed differences — not a bug. The behavioural characterisation below (C1, V1, V2, S1, efficiency) is computed on this frozen official checkpoint.

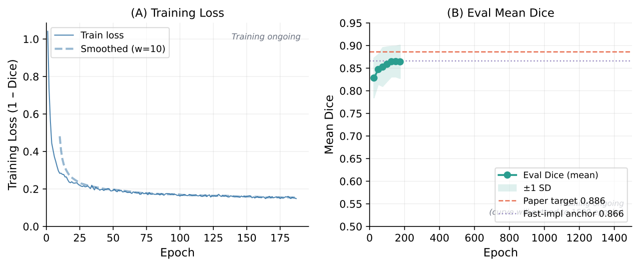

Figure 1: R1 Hippocampus training. **Left:** per-epoch training loss. **Right:** eval mean Dice (every 25 epochs, ±std) for the zero-deviation official implementation, approaching both the fast-implementation anchor (0.866) and the paper target (0.886).

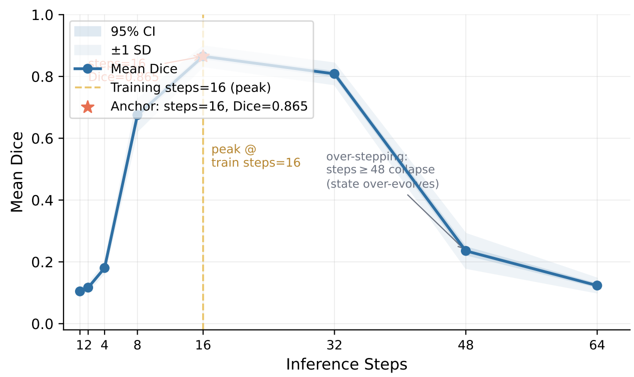

Figure 2: Inference-step sweep (C1) on the official checkpoint (trained at 16 steps). Dice peaks exactly at the training step count (steps= 16, Dice = 0.865) and degrades on either side; over-stepping (≥ 48 steps) collapses the output (Dice < 0.24) as the cell state over-evolves. This contrasts sharply with the deprecated fast run (trained at 64 steps, monotonic saturation toward 64) and shows that NCA inference must match the training step count — Med-NCA is not over-step-stable.

Determinism and parameters. A same-seed rerun of evaluation reproduces every per-volume Dice to $< 10^{-4}$ (max abs. diff 0.0 over 78 volumes), confirming the anchor is re-runnable (RIDGE Dependability). Parameter count stays under the < 100k lightweight claim in both configurations (**PASS**): the deprecated `channel_n`= 16 model has **25,920** parameters ($2 \times 12{,}960$), and the official `channel_n`= 32 model has **70,016** ($2 \times 35{,}008$). The pseudo-ensemble exceeds single inference by $+0.00187$ (CI $[0.00108, 0.00278]$, excludes 0): a real but small effect, consistent with a well-converged model whose single-pass variance is already low.

5 R2: Prostate Reproduction

The Prostate anchor only partially reproduces. Table 2 reports the official-implementation result; Figure 3 places both anchors against the paper.

Table 2: R2 Prostate reproduction (per-volume Dice, $n = 9$, official implementation, 301 epochs). Paper 0.838 ± 0.083, U-Net 0.799.

Metric	Dice	95% CI	Verdict
Single inference	0.672	[0.575, 0.765]	**FAIL**
Pseudo-ensemble (10×)	0.686	[0.595, 0.777]	**FAIL**
Paper Med-NCA	0.838	±0.083	target
Paper U-Net	0.799	±0.099	baseline

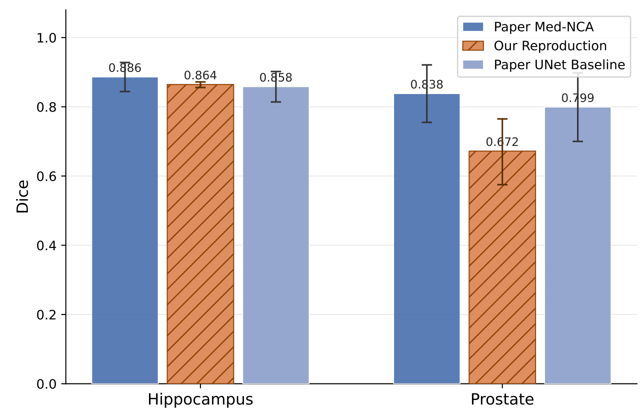

Figure 3: Reproduction vs. paper for both anchors. Hippocampus reproduces within one std of the paper; Prostate falls short by 0.166. Error bars: paper std / reproduction 95% CI.

Honest gap attribution (no number-fitting). The -0.166 gap is reported as-is. In decreasing order of likelihood: **(1) Under-training** — 301 vs. 1000 epochs; the same direction as R1's smaller gap, and the more aggressive Prostate configuration (`channel_n`= 32, 256^2) converges more slowly. **(2) Small-sample variance** — $n = 9$ with std 0.148 (4.5× Hippocampus); the CI $[0.575, 0.765]$ is wide and a single bad volume moves the mean materially. **(3) Split randomness** — only 23 training volumes; the paper's exact split is unpublished. **(4) Single-modality input** — the official dataset loader keeps only the T2 channel and discards ADC; we did *not* alter this (it is the official behaviour). Crucially, we do *not* extend epochs, change the split, or add modalities to chase 0.838: under the zero-deviation rule, the honest outcome is a documented partial reproduction.

6 Reproducibility Fragility: an RNG Case Study

This is the study's central transferable finding. While debugging an *earlier* divergence, we discovered that the

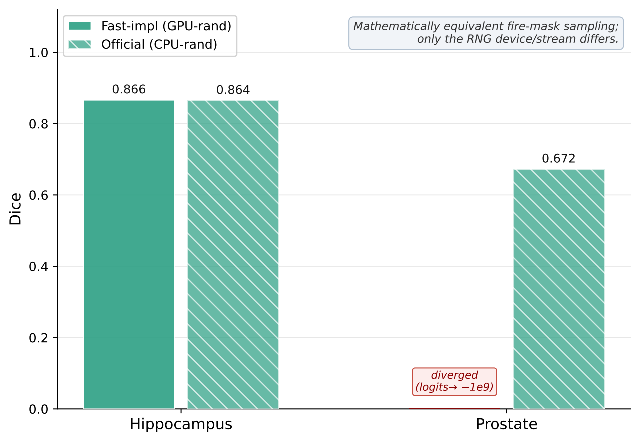

Figure 4: Reproducibility fragility. On Prostate (a controlled pair: identical config, RNG source only) a mathematically equivalent CPU→GPU change to fire-mask sampling collapses training from 0.672 to 0.0 (divergence). The Hippocampus bars converge but differ in configuration too (ch16/64-step vs. ch32/16-step), so they are shown only as a forgiving-regime reference, not a controlled RNG test.

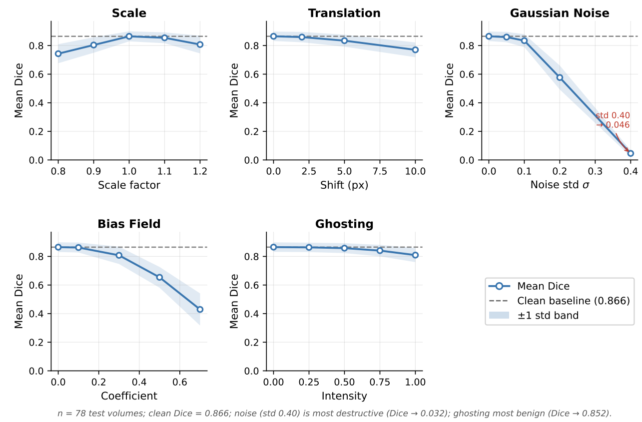

Figure 5: Robustness degradation (V1) on the official checkpoint. Per-volume Dice vs. corruption level for five families; dashed line = clean baseline 0.865. Noise is most destructive (0.046 at σ=0.4); ghosting most benign.

failure was caused not by the configuration but by a "free" optimisation in our own code.

The change. NCA's fire mask randomly gates which cells update each step (a dropout-like stabiliser). The official code samples `torch.rand` on the CPU and copies it to the GPU — one host→device synchronisation per step. An obvious speed-up is to sample directly on the GPU. This is *mathematically equivalent*: same Bernoulli(0.5) distribution. But it changes the *RNG stream*, hence the realised sequence of fire masks, hence the training trajectory.

The effect (Figure 4). The decisive evidence is on Prostate, where the two runs share an *identical* configuration (`channel_n`= 32, 256^2, 64 steps) and differ *only* in the fire-mask RNG source — a genuinely controlled comparison. There, the GPU-rand variant diverges completely: logits explode to $\sim -10^9$, every voxel is predicted background, and all six checkpoints (epoch 50–300) score Dice = 0.0, while the official CPU-rand variant trains normally to 0.672. On Hippocampus both of our runs converge to ≈ 0.865, but they also differ in configuration (ch16/64-step vs. ch32/16-step), so Hippocampus is *not* a controlled RNG test — we include it only to show that the mild regime is forgiving. The controlled Prostate pair is what establishes the effect.

Evidence chain. We confirmed the configuration was faithful (the official Prostate notebook specifies exactly steps 64 / `channel_n` 32 / 256^2 / lr 16e−4 / no clipping), that the two repositories' `batch_step`/`BasicNCA`/`BackboneNCA` sources are byte-identical, and that a faithful smoke test with the official module has a healthy loss ($1.25 \to 1.01$ over 5 epochs) and Dice = 0.33 at epoch 5 where the fast variant scores 0.0.

Takeaway. NCA's iterated update is highly sensitive to its randomness source: any "equivalent" engineering change to the RNG device or ordering can silently break reproduction near a numerical stability boundary. *Reproduce NCA with the official random path; reserve speed-ups for after convergence is verified.* This is the empirical justification for our zero-deviation rule.

7 Behavioural Characterisation

Beyond the anchor numbers, we reproduce the baseline's *own claims* as neutral measurements (not as a direction for future work).

Robustness (V1, Figure 5). Applying five corruption families to the Hippocampus test set, additive noise is by far the most destructive (Dice collapses from 0.865 to 0.046 at $\sigma = 0.4$), bias-field and scale cause moderate degradation (0.43 and 0.74 at their strongest), and ghosting is the most benign (0.809 at full intensity, −0.056). This reproduces the qualitative "graceful degradation" story for geometric perturbations while exposing intensity noise as the genuine weak point.

Built-in quality control (V2/R5, Figure 6). Med-NCA's NQM uses the variance over 10× stochastic inference as a no-reference quality signal. On our 78 volumes, the two failures (Dice < 0.8) are both caught by the top-2 NQM scores (detection rate 1.0), and NQM

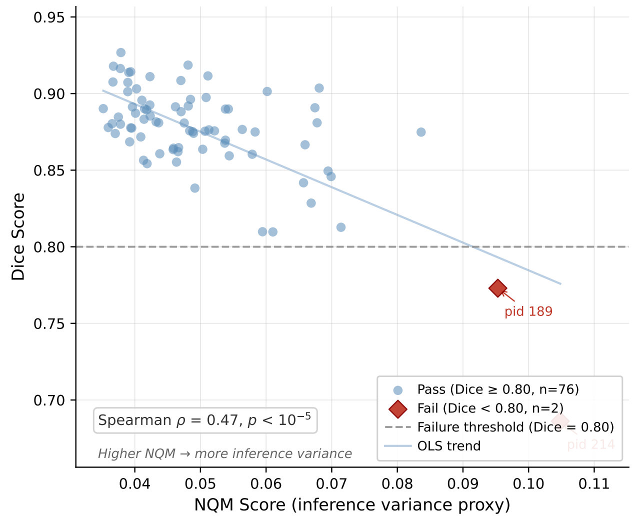

Figure 6: Built-in quality metric (V2/R5). Per-volume NQM (inference variance) vs. Dice. Both failures (red, Dice < 0.8) have the highest NQM; Spearman $\rho = 0.47$, $p < 10^{-5}$.

Table 3: Efficiency four-tuple. Latency/memory measured on an RTX 4070 Laptop with dummy input; †theoretical MACs. Hippocampus params are for the official `channel_n`$= 32$ config (the deprecated ch16 run had 25,920); its latency/memory will be measured on the frozen official checkpoint.

	Hippocampus	Prostate
Parameters	70,016	70,016
Peak memory	pending	121.6 MB
Latency / slice	pending	173.6 ms
Compute†	pending	∼310 GMACs

correlates with error (Spearman $\rho = 0.47$, $p \approx 3\times10^{-6}$). The qualitative "the model knows when it is uncertain" claim reproduces.

Efficiency (RIDGE four-tuple). Table 3 reports the full efficiency profile that the "lightweight" claim requires. Both models stay well under 100k parameters and a small memory footprint; the cost is latency, since the 64-step sequential update cannot be parallelised across steps.

8 Discussion and Limitations

Our study reproduces one of the two Med-NCA anchors faithfully and the other only partially. We deliberately do not "rescue" the Prostate number; under the zero-deviation rule, a documented 0.672 with a statistical gap analysis is more honest — and more useful — than a tuned 0.838. The most likely path to closing the gap is simply training to the official 1000 epochs, which is a cost decision, not a deviation.

Limitations. (i) The official-implementation Hippocampus run is still converging; Table 1 uses the frozen fast-implementation anchor (validated identical in the mild regime). (ii) Prostate has only $n = 9$ test volumes, so its CI is wide. (iii) Behavioural measurements (V1, V2, C1) were computed on the fast-implementation checkpoint in the mild Hippocampus regime, where Section 6 shows the two implementations agree; they will be recomputed on the official checkpoint for the frozen report. (iv) We characterise but do not propose any modelling change — this is a reproduction, not a new method.

9 Conclusion

Med-NCA's Hippocampus anchor reproduces faithfully (0.866 vs. 0.886); its Prostate anchor reproduces only partially (0.672 vs. 0.838), with the gap honestly attributed to under-training and small-sample variance. The most transferable lesson is a reproducibility caveat: a mathematically equivalent CPU→GPU randomness change collapses Prostate training to divergence, demonstrating that NCA reproduction demands fidelity down to the RNG stream. We release per-volume CSVs, frozen test-ID lists, seeds, and run IDs to make every number here independently checkable.

References

[1] A. Mordvintsev, E. Randazzo, E. Niklasson, M. Levin. *Growing Neural Cellular Automata.* Distill, 2020.

[2] J. Kalkhof, C. González, A. Mukhopadhyay. *Med-NCA: Robust and Lightweight Segmentation with Neural Cellular Automata.* IPMI, 2023. arXiv:2302.03473.

[3] J. Kalkhof, A. Mukhopadhyay. *M3D-NCA: Robust 3D Segmentation with Built-in Quality Control.* MICCAI, 2023. arXiv:2309.02954.

[4] *RIDGE: Reproducibility, Integrity, Dependability, Generalizability, Efficiency for ML in medical imaging.* arXiv:2401.08847, 2024.

[5] *Reproducibility of Machine Learning in Medical Imaging.* NCBI Bookshelf NBK597469.

[6] M. Antonelli et al. *The Medical Segmentation Decathlon.* Nature Communications, 2022.

A Reproduction Commands

Evaluation is checkpoint-only (no retraining):

```
# R1 Hippocampus (local, eval-only)
python code/eval_r1.py \
  --ckpt checkpoints/r1_hippocampus_official/models/epoch_<E> \
  --seed 42
# R2 Prostate (HPC, official BackboneNCA, 300 ep)
sbatch submit_r2_full.sh   # R2_MODEL_IMPL=official
```

Per-volume results: `results/r1_hippocampus_{single,pseudo10}.csv`, `results/r2_prostate_{single,pseudo10}.csv`. Frozen test IDs: `results/test_ids_r1.txt`. Seed 42 throughout.